

Wavelet-Enhanced Deep Learning Ensemble for Accurate Stock Market Forecasting: A Case Study of Nifty 50 Index



Course : Machine Learning(CS308)

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Problem Statement

- Stock data is highly **unstable, noisy, and hard to predict**.
- **Single models** (LSTM, CNN, TCN) fail to capture all patterns.
- **Noise** in data lowers prediction accuracy.
- Past studies use **only one model** or **only denoising**, not both.
- No existing system combines **wavelet denoising + multiple deep models**.
- **Main Problem** : Build a **robust ensemble model** using **denoised data + deep learning models** for better prediction.

MOTIVATION

- Traditional models fail to capture complex patterns.
- Financial data is highly noisy and dynamic.
- Need for hybrid intelligent models.

Importance:

- Helps investors in decision-making.
- Reduces financial risks.
- Enables better forecasting.

LITERATURE REVIEW

- **Existing Approaches :**

- ARIMA (Statistical model)
- LSTM (Deep Learning)
- CNN-based models

- **Limitations :**

- Poor noise handling
- Overfitting issues
- Limited feature extraction

Proposed Approach :

Wavelet + Deep Learning Ensemble improves prediction accuracy.

Dataset Description

- **Features and Data Types:**
- **Start Date:** 2010-01-04
- **End Date:** 2025-12-29
- **Total Observations:** 3434
- **Total Attributes:** 4
- **Data Type:** Time-Series (Tabular)

Feature	Type
Open Price	Numerical
High Price	Numerical
Low Price	Numerical
Close Price	Numerical

- **Target Variable:** Close Price (Numerical)
- Real-world **highly volatile** stock index data
- Contains **noise**
- **Wavelet (Coif3) denoising** applied

Methodology

Step 1: Data Collection.

Step 2: Data Preprocessing.

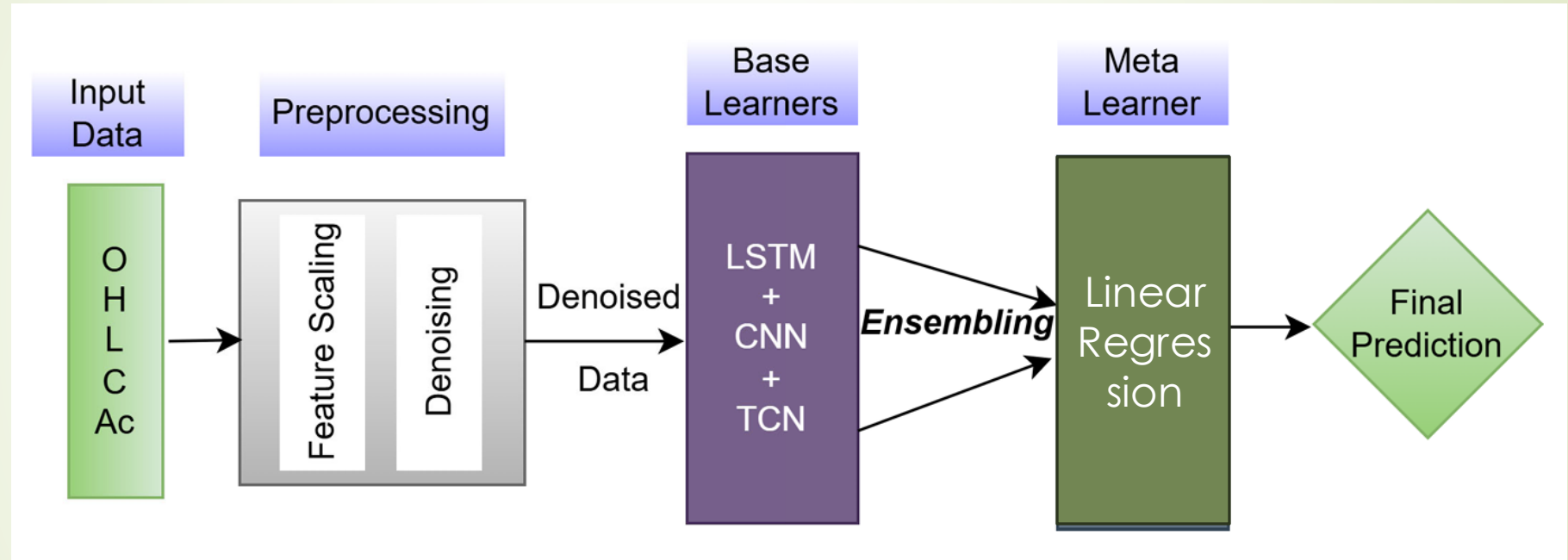
Step 3: Wavelet Transform (Noise Reduction).

Step 4: Model Training (LSTM).

Step 5: Ensemble Learning.

Step 6: Prediction & Evaluation.

System Architecture

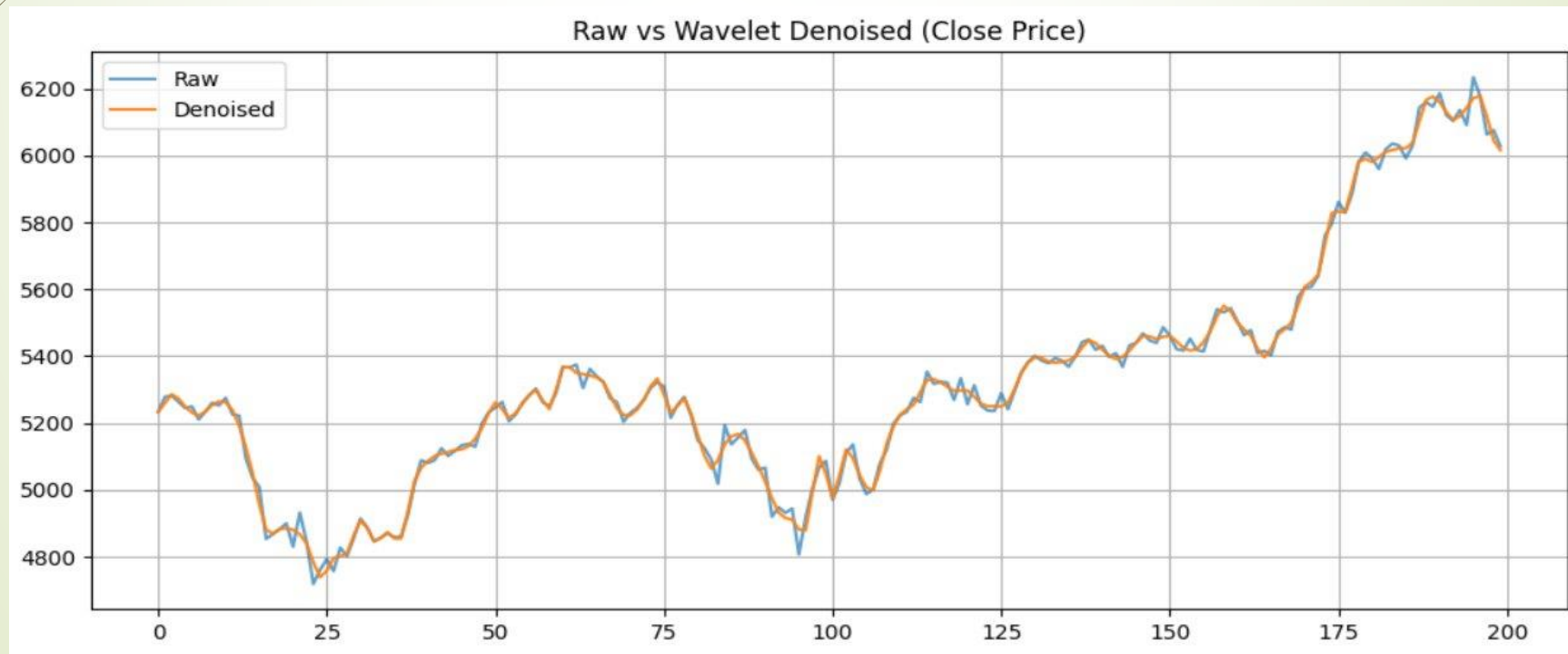


WAVELET TRANSFORM

- Used for signal decomposition.
- Removes noise from stock data.
- Improves feature extraction.

Benefit:

Enhances model accuracy and stability.



MODEL USED

Model:

- LSTM (for time-series learning)
- CNN (for feature extraction)
- TCN (Temporal Convolution Network)
- Ensemble Model (Final prediction)

Why LSTM?

- Captures time-series dependencies.
- Handles sequential data effectively.

Why Ensemble?

- Combines strengths of all models,
- Gives best accuracy.

IMPLEMENTATION

Implementation Details:

- Language: Python
- Libraries: NumPy, Pandas, TensorFlow, Keras, Matplotlib

Steps:

- Data Loading.
- Data Preprocessing.
- Model Training.
- Prediction.

1.2 Import Libraries

```
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt
import math
import random
import os
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
from sklearn.linear_model import LinearRegression

import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import (
    Input, LSTM, Dense, Dropout, Conv1D, MaxPooling1D,
    Flatten, Add, Activation, GlobalAveragePooling1D
)
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping
```

1.3 Set Random Seed

```
np.random.seed(42)
tf.random.set_seed(42)
```

1.4 Download Stock Data

```
df = yf.download("^NSEI", start="2010-01-01", end="2023-12-31").dropna()
df = df[["Open", "High", "Low", "Close"]]

/tmp/ipykernel_8471/4121186968.py:1: FutureWarning: YF.download() has changed argument auto_adjust default to True
df = yf.download("^NSEI", start="2010-01-01", end="2023-12-31").dropna()

[*****] 1 of 1 completed
```

1.5 Save Dataset to Drive

```
df.to_csv(f"{project_path}/nifty50_raw.csv")
```

1.6 Keep RAW Copy

```
df_raw = df.copy()
```

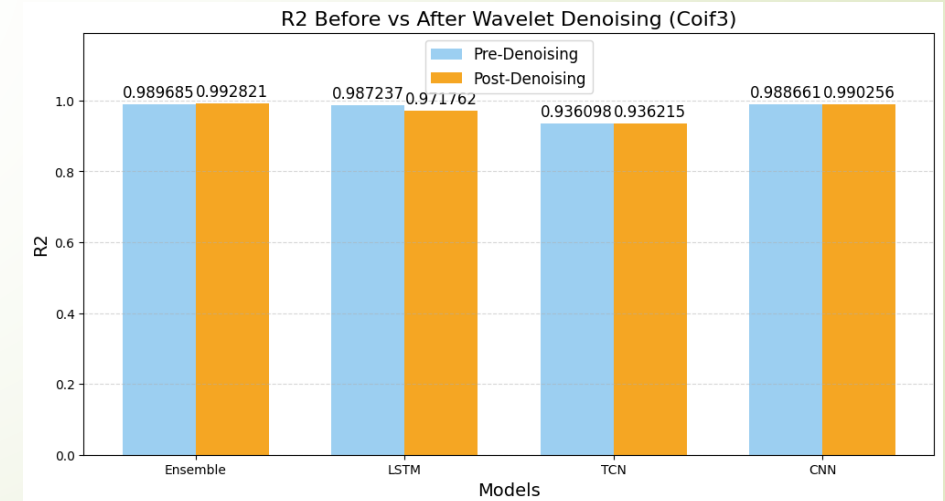
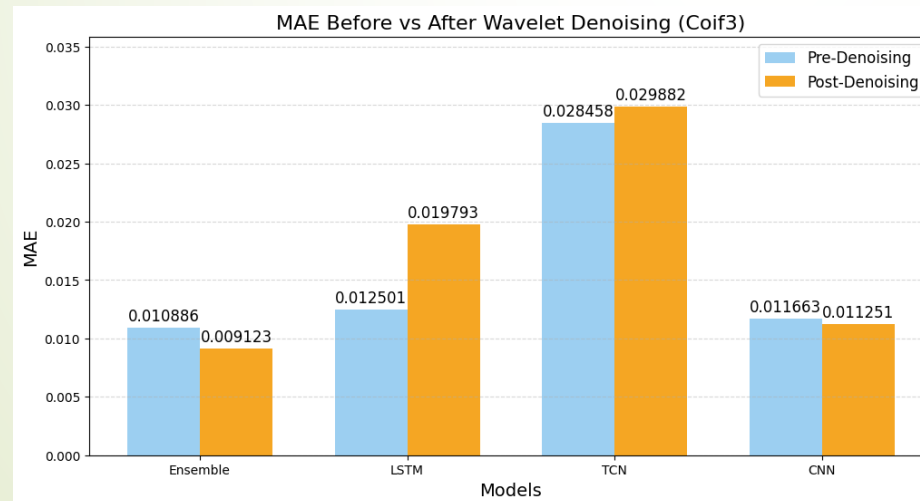
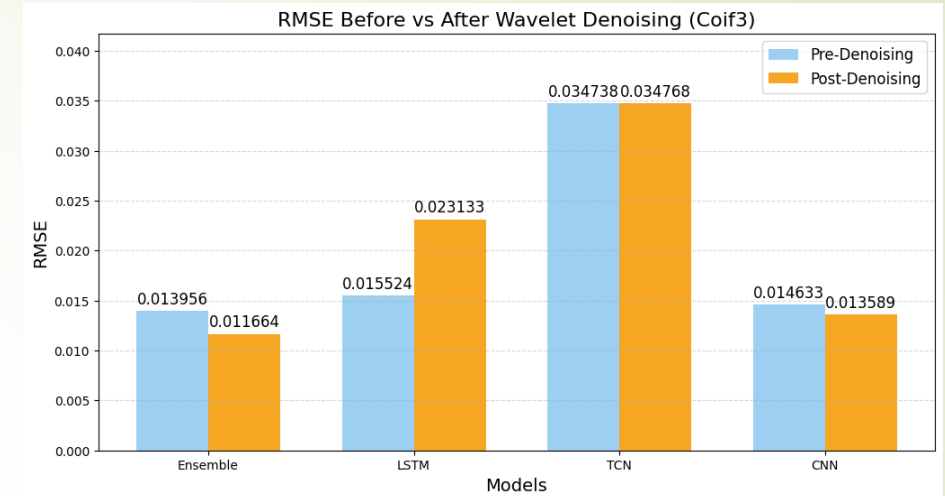
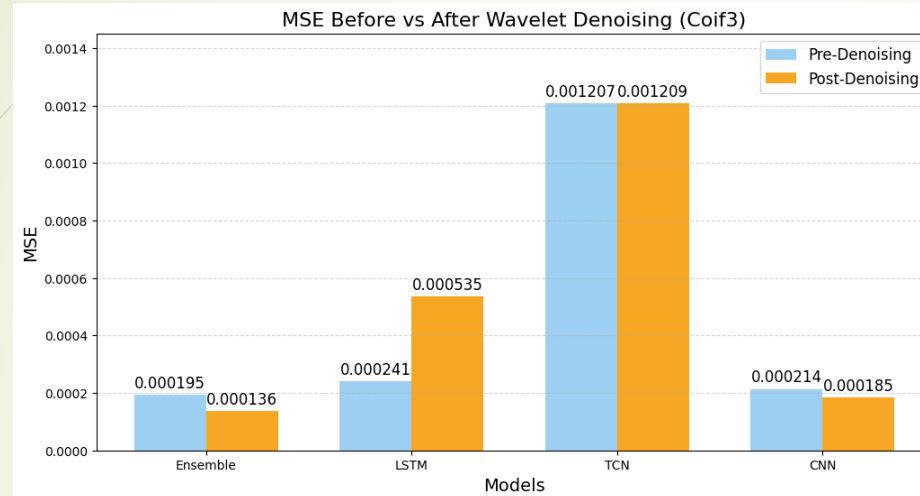
1.7 Preview Data

```
print(df.head())
print("Shape:", df.shape)
```

Price Ticker	Open ^NSEI	High ^NSEI	Low ^NSEI	Close ^NSEI
Date				
2010-01-04	5200.899902	5238.450195	5167.100098	5232.200195
2010-01-05	5277.149902	5288.350098	5242.399902	5277.899902
2010-01-06	5278.149902	5310.850098	5260.049805	5281.799805
2010-01-07	5281.799805	5302.549805	5244.750000	5263.100098
2010-01-08	5264.250000	5276.750000	5234.700195	5244.750000

Shape: (3434, 4)

Results

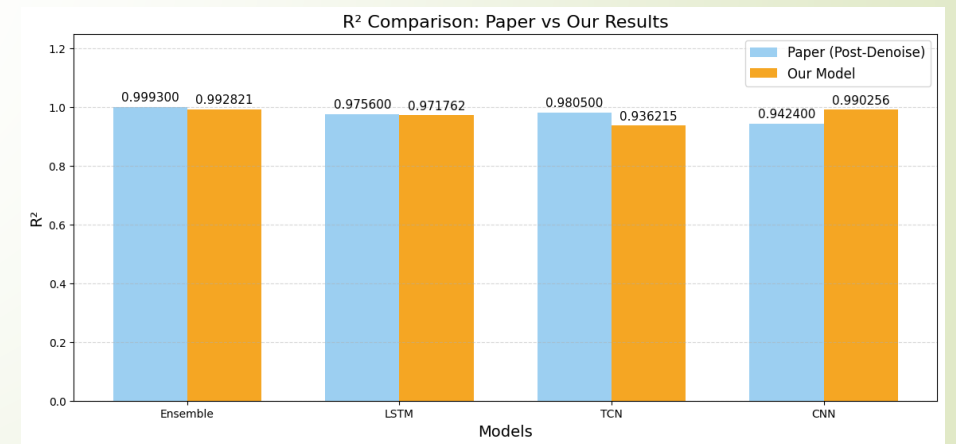
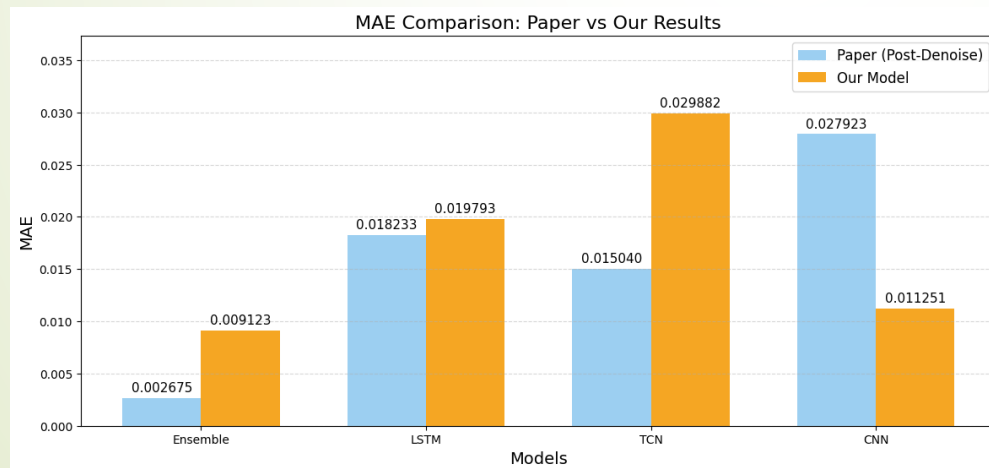
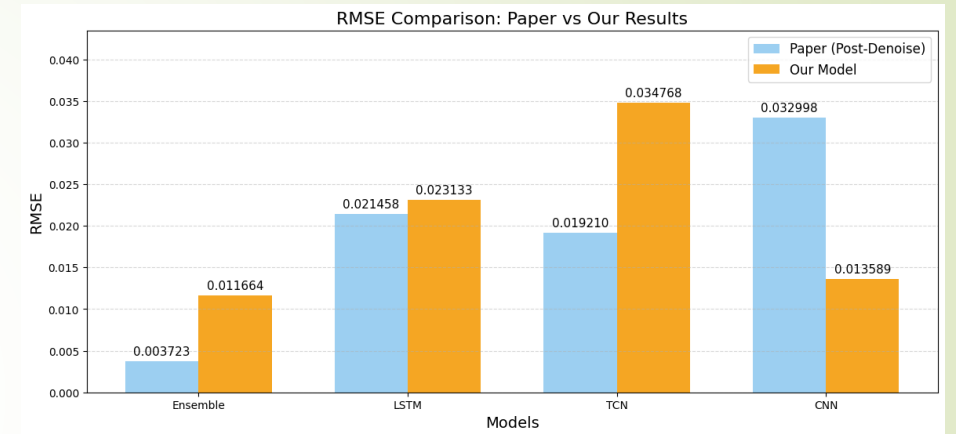
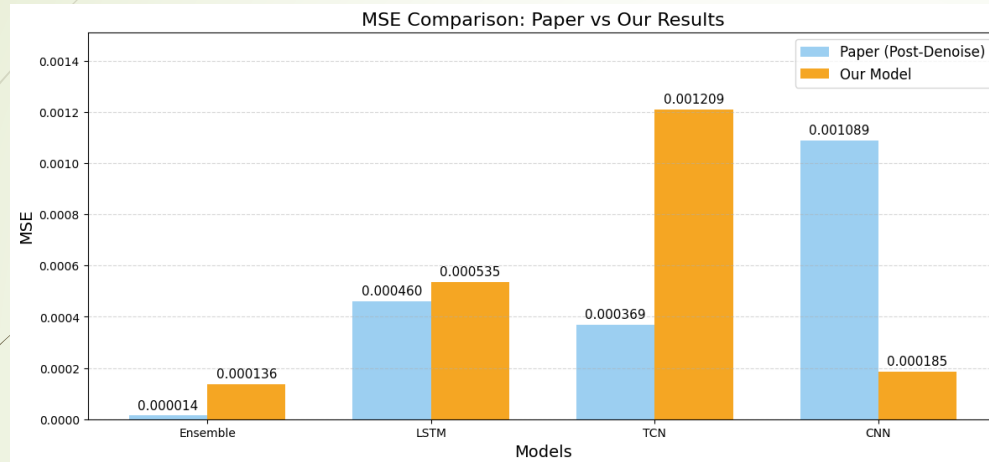


Delta changes

- Noise is present in all features, not just Close
- Cleaning all inputs gives a clearer signal and better model learning
- Our model predictions are **smooth**, so LR combines them better than other models
- LR is **more stable** and avoids overfitting on denoised data

Category	Paper	Our Code
Denoising	Coif3 on Close only	Coif3 on all OHLC
Ensemble	Random Forest Stacking	Linear Regression

Delta changes Results



PERFORMANCE COMPARISON

Comparison with Traditional Models:

Traditional Models:

- Less accurate.
- Sensitive to noise.

Proposed Model:

Higher accuracy.

Better noise handling.

Stable predictions.

Best Model:

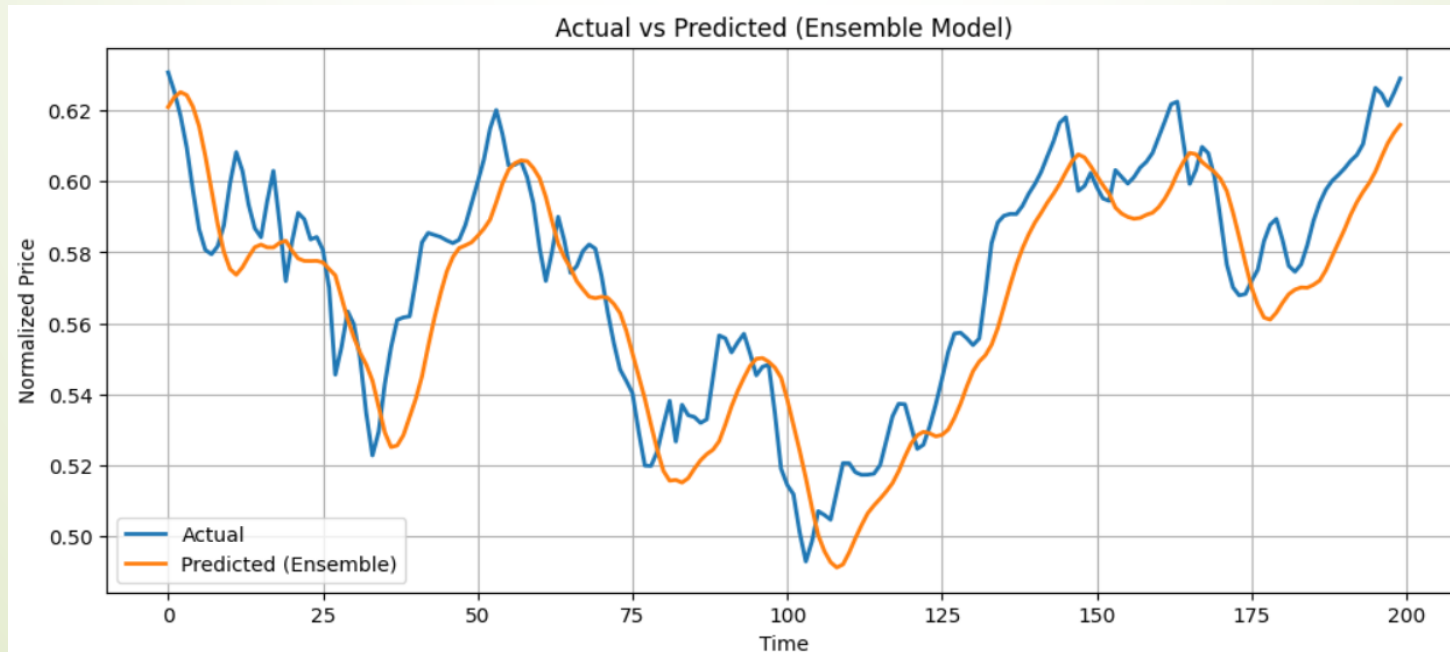
- LSTM gives best performance.
- Ensemble provides stable results.

CONCLUSION

Wavelet-Enhanced Deep Learning Ensemble significantly improves stock price prediction.

Applications:

- Financial forecasting.
- Investment decision-making.
- Risk management.



REFERENCES

- IEEE Research Paper:

<https://ieeexplore.ieee.org/document/11005722>

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- Python Libraries Documentation.
- Yahoo finance Nifty 50 dataset source.